

STAD29 / STA 1007 assignment 6

Due Tuesday Mar 19 at 11:59pm on Quercus

Packages for this one:

```
library(MASS)
library(car) # for Manova later, for me

## Loading required package: carData

library(tidyverse)

## -- Attaching packages ----- tidyverse 1.2.1 --
## v ggplot2 3.1.0      v purrr 0.3.1
## v tibble 2.0.1      v dplyr 0.8.0.1
## v tidyr 0.8.3       v stringr 1.4.0
## v readr 1.3.1      v forcats 0.3.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## x dplyr::recode()  masks car::recode()
## x dplyr::select() masks MASS::select()
## x purrr::some()   masks car::some()
(I use car later, but you probably won't.)
Hand in problem 2.
```

1. Work through Chapter 22 of PASIAS.
2. A survey called High School and Beyond was given to a large number of American high school seniors (grade 12) by the National Center of Education Statistics. The data set at <http://www.utsc.utoronto.ca/~butler/d29/hsb.csv> is a random sample of 200 of those students.

The variables collected are:

- **gender**: student's gender, female or male.
- **race**: the student's race (African-American, Asian,¹ Hispanic, White).
- **ses**: Socio-economic status of student's family (low, middle, or high)
- **scht**: School type, public or private.
- **prog**: Student's program, general, academic, or vocational.
- **read**: Score on standardized reading test.
- **write**: Score on standardized writing test.
- **math**: Score on standardized math test.
- **science**: Score on standardized science test.
- **socst**: Score on standardized social studies test.

We have seen these data before. This time, our angle is to explore any relationship between socio-economic status and the five test scores at the bottom of the list, using discriminant analysis.

- (a) (2 marks) Read in and display (some of) the data.
- (b) (2 marks) Run a discriminant analysis to assess association between socio-economic status and the five test scores `read` through `socst`. Display the results.
- (c) (2 marks) How many of the linear discriminants do you think are worth looking at? Explain briefly.
- (d) (3 marks) Which *three* of the test scores contribute the most to LD1? Explain briefly. (If you think that fewer than three test scores contribute to LD1 at all, name the ones you think are important and justify briefly your choice of fewer than three.)
- (e) (3 marks) What would make a student have a *high* score on LD1, based on what you said in the previous part?
- (f) (3 marks) Obtain the “predictions” for your discriminant analysis and make a data frame which is the original data plus the output from the predictions, side by side. Display (some of) the results.
- (g) (3 marks) Plot the LD1 scores against the LD2 scores for each student, colouring each point by which socio-economic status group the student was from. Do the groups appear to be distinct, or not? Explain briefly.
- (h) (2 marks) Create a new variable called `is_correct` in your data frame that is the text `correct` if the predicted SES group is the same as the observed one, and `wrong` otherwise. Save your new data frame.
- (i) (3 marks) Modify your graph of part (g) to include information about whether each student was correctly classified or not. You can use the shape of each point to encode this. (You can use the idea in problem 20.3(f) of PASIAS.)
- (j) (3 marks) For each SES group, where on your graph are the ones that are correctly classified?
- (k) (2 marks) Obtain a table showing how many students were classified into each combination of observed and predicted `ses` category.
- (l) (2 marks) Do you think the classification is good or bad? Comment briefly.
- (m) (3 marks) Find the overall proportion of students who got misclassified. (Note that you can use your `is_correct` that you obtained earlier.)
- (n) (3 marks) Find the proportion of students from each (observed) socio-economic group who got misclassified. Does one group seem to be easier to classify than the others? Explain briefly.

Notes

¹I’m always amused at how Americans put all Asians into one group.

²Which is actually the last column, confusingly.